

Moorthy Chetan

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SUMMARY

Third-year B.Tech Information Technology student focused on software engineering, backend systems, and applied AI. Experienced in building real-time applications, zero-knowledge systems, and resource-aware local AI infrastructure using Java, Python, JavaScript, Node.js, React, and PostgreSQL.

Technical Skills

Languages: Java, Python, JavaScript, SQL

Backend: Node.js, Express, REST APIs, Socket.IO

Frontend: React, Next.js

Databases: PostgreSQL, MongoDB, SQLite

AI / Systems: llama.cpp, sentence-transformers, Local LLMs

Security / Tools: Web Crypto API, Git

Projects

LocalHarness – Multi-Agent Local AI Orchestration *Python, llama.cpp, SQLite, sentence-transformers*

- Built a hierarchical **10-agent AI system** with a lean 4-tool root orchestrator for coding, research, data engineering, document analysis, career, and system workflows.
- Designed a **3-tier resource-aware model architecture** that dynamically routes tasks across lightweight, general-purpose, and heavy reasoning models on a 16 GB RAM laptop with Radeon 780M integrated graphics.
- Implemented **tool capability isolation** preventing agents with untrusted web-ingestion tools from simultaneously accessing host-dangerous execution capabilities.
- Added **SQLite-backed semantic memory** using local MiniLM embeddings and chunked document inspection for long-context workflows.
- Built model lifecycle controls for graceful shutdown, unified-memory reclamation, health checks, endpoint synchronization, and automatic task-based model switching.

PrivChat – End-to-End Encrypted Real-Time Chat *Next.js, Node.js, Socket.IO, PostgreSQL, Web Crypto API*

- Designed a **zero-knowledge messaging architecture** where encryption and decryption occur entirely in the browser using per-message AES-GCM encryption and RSA-OAEP key wrapping.
- Generated and persisted **RSA-4096 keypairs in IndexedDB**, synchronizing only public keys while retaining private keys exclusively on the client.
- Implemented real-time encrypted message delivery using **Socket.IO** with PostgreSQL persistence for ciphertext, IVs, and per-recipient encrypted message keys.
- Integrated Google OAuth authentication using Passport and structured the application with Next.js and a Node.js/Express backend.

FocusSync – Real-Time Collaborative Study Platform *React, Node.js, Express, Socket.IO, MongoDB*

- Built a **server-authoritative real-time focus timer** using Socket.IO with synchronized state and late-join session recovery.
- Implemented browser-based distraction tracking using the **Page Visibility API** and a dynamic focus-scoring system based on distractions and session completion.
- Developed REST APIs and analytics dashboards for **session history, statistics, leaderboards, and daily focus streaks**.
- Structured authentication, session persistence, and real-time room coordination using Node.js/Express and MongoDB with Mongoose.

Education

Vellore Institute of Technology, Vellore

B.Tech in Information Technology

Class XII

95.3%

Class X

96.4%

2024 – 2028

CGPA: 8.34

2024

2022